

Linux Interview Question Scenario-Based

1. you need to find out why a server is running slowly. what steps you would you take?

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--> start by checking the system load with top or htop

top

htop

--> look at CPU and memory usage, and identify any processes consuming excessive resources

--> use iostat and itop to check disk I/O

iostat

itop

--> free -m for memory usage, and df -h for disk space

free -m

df -h

--> check for any network issue using netstat or ss

netstat

ss

--> finally, review system logs in /var/log for any error or warnings

/var/log

2. a user's home directory is filling up disk space on the root partition. how would you resolve this?

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--> check the disk usage with du -sh /home/user to identify large files or directories

du -sh /home/user

--> if possible, move large files to another partition with more space using mv or rsync

mv

rsync

--> alternatively, advise the user to delete or compress(tar, gzip) unnecessary files

3. you need to secure a web server against common vulnerabilities. what measures would you take?

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- > ensure all software is up-to-date using the package manager (apt, yum)
- > configure a firewall using iptables or firewalld
- > implement HTTPS using SSL/TLS certificates (via lets encrypt or another CA)
- > use tools like fail2ban to block malicious IPs and configure SELinux or AppArmor for enhances security

4. The SSH service on your server is being targeted by brute force attacks. what steps would you take to mitigate this?

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- > change the default SSH port from 22 to something less common in /etc/ssh/sshd_config
- > implement key-based authentication and disable password authentication
- > use fail2ban to block IPs after a certain number of failed login attempts
- > allow only specific IP addresses to connect using iptables or ufw

5. you need to automate backups of a directory to a remote server. how would you do it?

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- > using rsync for efficient file transfers
- > set up SSH key-based authentication between the local and remote servers
- > write a script to run rsync with the desired options
- > schedule it using cron by adding cron job (crontab -e)

6. a critical service has crashed and won't restart. what steps do you take to troubleshoot and resolve the issue?

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- > check the status and logs of the service using `systemctl status servicename` and `journalctl -u servicename`
- `systemctl status servicename`
- `journalctl -u servicename`

--> look for any configuration errors or missing dependencies of the service

--> verify the configuration file syntax

nginx -t for nginx

apachectl configtest for apachectl

--> resolve any issues found and restart the service with

systemctl restart servicename

7. your servers time is out of sync, causing issues with applications. how do you fix this?

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--> install and configure ntpd and chrony to synchronize time with NTP servers

ntpd

chrony

--> use ntpq -p to check the status of NTP peers

ntpq -p

--> ensure the correct time zone is set using timedatectl

timedatectl

8. you need to create a new user and ensure they have no shell access. how would you do this?

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--> use the useradd command with the -s option to set the user's shell to /sbin/nologin or /bin/false

example: useradd -s /sbin/nologin username

9. your server is running out of memory and starting to swap heavily. what actions would you take?

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--> identify memory-hogging processes using top or ps aux --sort =-%mem

top

ps aux --sort =-%mem

--> consider stopping or restarting these processes

--> increase swap space by creating a swap file if needed

dd if=/dev/zero of=/swapfile bs=1G count=4, mkswap/swapfile, swapon/swapfile

--> optimize application memory usage or ass more physical memory to the server

10. you need to restrict a user's disk usage. how would you implement this?

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--> use disk quotas

--> first enable quotas on the file system by modifying /etc/fstab and remounting the file system

--> then use edquota to set user-specific disk quotas and

--> quotaon to enable quotas

--> check usage with quota

11. a service needs to start on boot. how do you ensure this?

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--> enable the service using systemctl enable servicename

systemctl enable servicename

--> this creates the necessary symlinks for the service to start on boot

--> verify with systemctl is-enabled servicename

systemctl is-enabled servicename

12. your web server is showing a 502 bad gateway error. what steps do you take to troubleshoot?

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--> check the status and logs of both the web server (eg: Nginx) and the backend service it proxies to (eg: PHP-FPM)

--> verify the backend service is running and reachable. check configuration files for any misconfigurations

--> ensure sufficient resources (memory, CPU) are available

13. you need to schedule a recurring task to clean temporary files. how would you do this?

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--> write a script to clean temporary files

```
#!/bin/bash\nrm -rf /tmp/*
```

--> make it executable

```
chmod +x scriptname.sh
```

--> add a cron job to execute this script at desired intervals using crontab -e

eg: 0 0 * * * /path/to/scriptname.sh

14. you need to compile and install software from source. what steps do you follow?

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--> download the source code tarball

--> extract it using tar -xzf filename.tar.gz

--> change into the extracted directory

--> read the README or INSTALL file for specific instructions

--> typically, run ./config, make and make install command for install packages from source code

15. you need to find and kill all processes started by a specific user. how do you do this?

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--> use the ps command to list processes by the user (ps -u username)

```
ps
```

```
ps -u username
```

--> pipe this to grep and awk to extract PIDs

--> then pass them to xargs to kill

```
ps -u username | grep -v PID | awk '{print $1}' | xargs kill -9
```